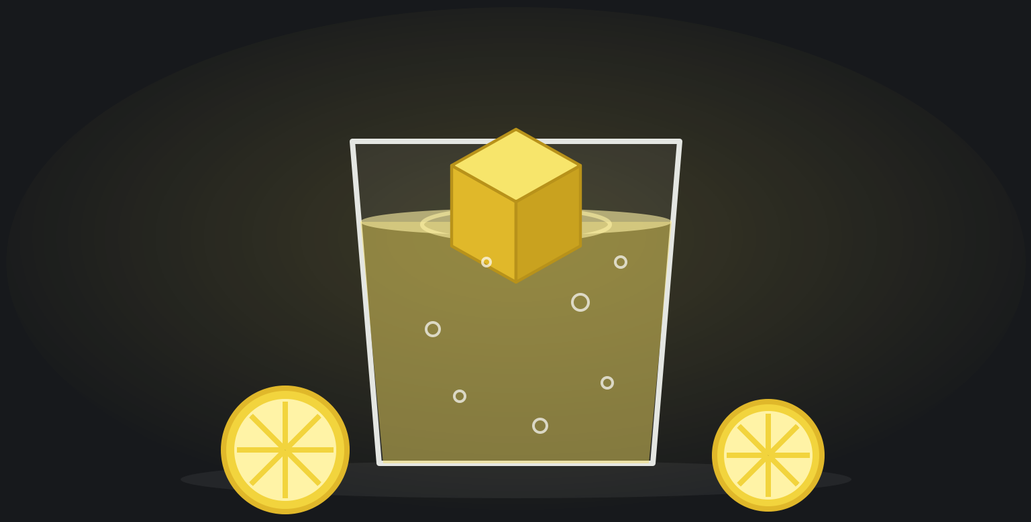
PROPOSAL FOR THE SINGHA ULTIMATE CHALLENGE 2026

Lemon Soda

One dissolvable cube makes a lemon drink in the glass

October 2026



Illustration, not packaging artwork.

Executive Summary

1 Problem statement

- Lemon drinks are sold ready to drink in cans.
- No domestic dissolvable drink cube has been found.
- The gap is a new form of drinking, not a new flavour.

2 Identity problems

- Singha already lists a lemon soda can, which its product page says has no sugar.
- So Lemon Soda overlaps its flavour.
- The name needs a register search (Gate 7). The lion and SINGHA are for this proposal only.

Lemon Soda

One cube, dissolved in one 330 ml serve, made in the glass

3 Proposal

- One cube, dissolved in one 330 ml serve; one cube per pouch, not yet confirmed by Singha.
- Desk case: THB 4.91 contribution per cube.
- NPV THB 0.88 million at 200,000 cubes a year, THB 500,000 fixed cost (page 13).

4 KPI and expected goal

- 3 units per store per week; repeat 30%, floor 15% (all assumptions).
- 60% sell-through in 90 days; two listings in six months.
- Goal: a measured go or stop decision at M10 to M11.

The ask: a gated proof programme, not a launch: a go decision, named gate owners and twelve decisions (page 17).

Desk model on assumed volumes and fixed costs, not a forecast. Sources: appendix C.

WHY THIS PRODUCT

Singha already lists a lemon soda; Lemon Soda competes with it on form, not flavour

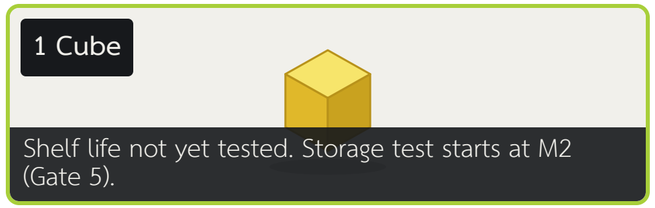
SINGHA LEMON SODA CAN (LISTED ON THE COMPANY PAGE)

Form	Ready to drink
Pack	330 ml can
Sweetener	Sucralose and acesulfame potassium (company page)
Claim	Company says no sugar (its own claim, not checked against a raw save)
Price	Not verified

LEMON SODA, PROPOSAL

Form	Dissolves in 330 ml of water, made in the glass
Pack	One cube per single-serve pouch
Sweetener	Sucralose, about 100 mg/kg (assumption)
Claim	No sugar-free, health or sodium claim
Price	THB 15 base case (assumption)

> Flavour overlap is close; the company says its lemon soda has no sugar. Cannibalisation is one of several commercial risks (Phase 5).



Sources: Singha Lemon Soda product page, extract (S); Thai retail listings not dated. Overseas cubes: Ricola (P). Stage 1 is an illustration, not packaging.

CONSUMPTION FORM

The consumer makes the drink in the glass: one cube dissolved in 330 ml of water

1 Open the pouch

One single-serve pouch holds one cube.

2 Drop the cube into 330 ml of water

The cube is designed at 14 mm and about 3.8 g.

3 Stir, then drink

Working pH 3.5. Taste is not yet tested.

> The stimulus stays withheld until the brand and legal rulings are in (Gate 7).

FRONT LABEL, DRAFT WORDING

One compressed cube is dropped into 330 ml of water.

Serving: 1 cube made up to 330 ml.

Contains an intensive sweetener: sucralose.

Single-serve pouch. Net weight: 3.8 g (specification; verified by weighing before print).

Wording only, not cleared. No vitamin, sugar-free or health statement is printed.



Label wording is a draft and not cleared. Stage 2 is an illustrative diagram.

SERVE COMPOSITION

One serve needs 0.55 volume of CO_2, 0.81 g potassium bicarbonate and 33 mg sucralose

0.55 vol

CO_2 per 330 ml serve (design value; R&D sets the release gate)

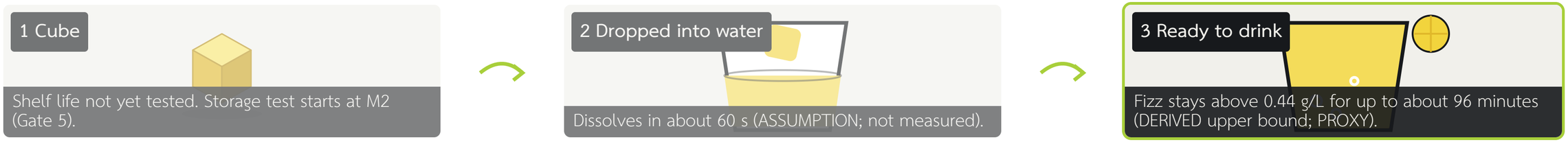
0.81 g

Potassium bicarbonate, about 8.1 mmol of CO_2

33 mg

Sucralose, about 100 mg/kg (DERIVED); category limit 300 mg/kg, pending Gate 1

Citric acid	2.10 g in total, of which 0.52 g neutralises the bicarbonate; the rest sets pH 3.5
Binder and filler	0.80 g (assumption; not yet named or checked for sodium)
Lemon flavour	0.05 g (assumption)
Potassium	317 mg per serve, about 9% of a Thai reference of 3,500 mg a day; the desk model does not cover people on potassium-retaining medicines or with kidney disease
Working pH	3.5 (anchor only; taste is untested)



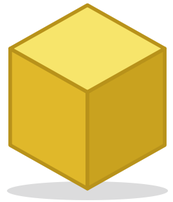
Derived from stoichiometry and the desk model. Reference values are secondary copies (appendix C). Citric acid pKa values are UNVERIFIED. Nothing has been measured.

CUBE AND POUCH

The cube's specification is 14 mm and 3.8 g; net weight is printed after weighing (Gate 9)



Approved pack artwork, cropped to the box and cube.



14 mm side (illustration)

Cube side	14 mm
Central dry-cube mass	3.79 g, range 3.46 to 4.24 g (derived, not weighed)
Density	1.38 g/cm³, band 1.25 to 1.60 g/cm³ (assumption)
Packed unit	5.29 g with a 1.5 g pouch (assumption)
Sweetener	Sucralose, about 33 mg per serve (about 100 mg/kg, assumption)
Net weight on pack	3.8 g specification; verified by weighing before print (Gate 9)

Derived on assumed inputs. Nothing has been weighed yet; net weight is confirmed from weighed samples before print (Gate 9).

PACK AND CONTAINER

A six-cube carton fits about 128,600 cubes in a 20 ft container

6
cubes per carton (recommended)

21,431
cartons per 20 ft container

128,586
cubes per 20 ft container

2.82 t
modelled load, against a 28.3 t maximum payload

Larger option: twelve cubes

128,580 cubes in 10,715 cartons per 20 ft container. Carton 190 × 150 × 100 mm (assumption).

Recommended carton

190 × 150 × 50 mm, printed folding paperboard (assumption). Six flat pouches; 40 mm of the 50 mm height used (assumption, not a packing trial).

Sources: container specifications, ONE and SeaCube (S). Carton size and packing share are assumptions; the counts are not a load plan.

COMPETITIVE POSITION

The Singha Lemon Soda can and Lemon Soda share a flavour and differ in form and pack

	Singha Lemon Soda can	Lemon Soda
Flavour	Lemon	Lemon
Form	Ready to drink	Dissolves in 330 ml of water, made in the glass
Pack	330 ml can	One cube per single-serve pouch
Sweetener	Sucralose and acesulfame potassium (company page)	Sucralose, about 100 mg/kg (assumption)
Shelf price	Not verified	THB 15 base case (assumption)
Claims	Company says no sugar (company's claim)	No sugar-free or health claim
Status	Listed on company page (extract); retail sale not dated	Desk model only; no test has run

> The lemon soda is the closest substitute, and the company says it has no sugar. Cannibalisation is reviewed at Phase 5 when 50% or more of buyers say they switched.

Sources: Singha Lemon Soda product page, extract (S). Lemon Soda's prices are assumptions; the lemon soda price is not verified.

EXTENSION OPTIONS

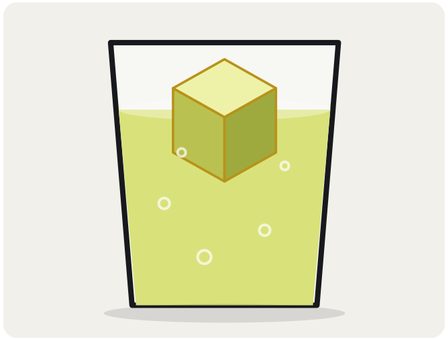
The same drinking way could carry Singha's other soda flavours; only lemon is in scope



Lemon

In scope: this proposal

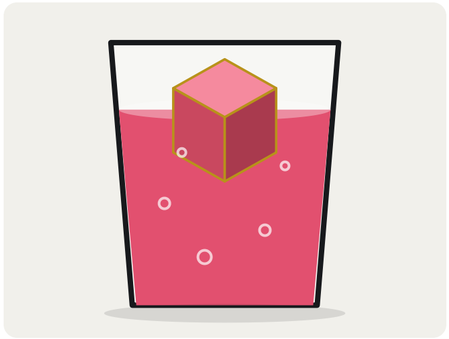
Company page says no sugar.



Yuzu

Option, not in scope

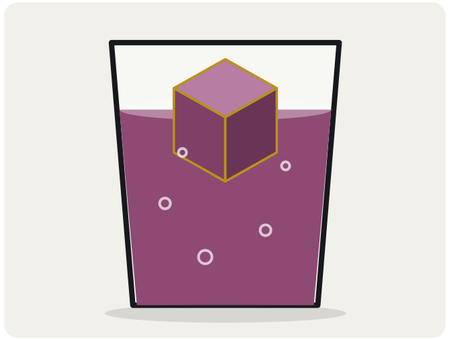
Company page says no sugar.



Red lemon

Option, not in scope

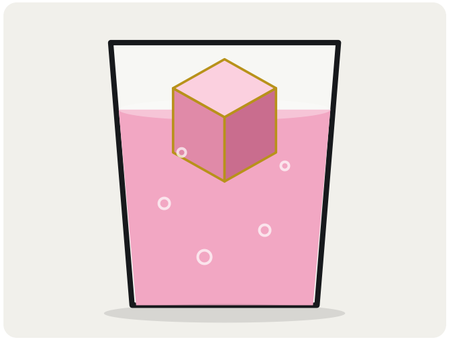
Company page says no sugar.



Ume

Option, not in scope

No sugar statement on the page.



Pink

Option, not in scope

No sugar statement on the page.

> The drinking way is the same in each: one cube in one 330 ml serve. Options are for Singha to weigh after the lemon proof. Each needs its own taste test, sweetener check and label clearance; none is tested or claimed here.

Flavour names and sugar wording: Singha Lemon Soda product page, extract (S). Cube colours are illustrative, not product colours.

TARGET SEGMENT

Men aged 25 to 34 are the proposed lead; the desk gap over men aged 35 to 44 includes zero

0.5 to 1.5

points: median desk gap between men aged 25 to 34 and men aged 35 to 44

−0.03 to 1.39

points: 90% interval at a 30% baseline, so it includes zero

- The lead is a proposal decision, not a consumer finding.
- The desk model rests on drinking of sugar-sweetened drinks. The cube contains no sugar, so transfer to a sweetener-only product is untested.
- The survey does not split ages 25 to 34 from 35 to 44; the split is modelled.
- Men aged 45 to 59 are a later volume segment and are pended until the lead is read.

Desk model on NSO survey inputs. Printed survey pages are not yet checked (UNVERIFIED).

MARKETING CAMPAIGN

The campaign tests direction and repeat before any paid media is scaled

M3 to M5

Concept pre-test

- Recruited online panel: men aged 25 to 59 who drank alcohol in the past year
- One brand-blind concept, signed off by M3
- Top-two-box purchase intent by age cell, at one team price (THB 15, assumption)
- Rule fixed in advance; it reads direction, not size
- No media, no sale

M7 to M10 · 13 weeks

Pilot, two routes

- Step 1: click-to-purchase read on non-paid traffic, first four weeks
- Route A: in-store trial in pilot stores; at least 3 units per store per week
- Route B: paid social to one listing, only after Singha sets the cost threshold; at base inputs it does not recover its first-purchase cost in one serve (DERIVED)
- Repeat read at 8 and 12 weeks

M10 to M11

Measured decision

- Scale one route, repeat once in the same stores, or stop
- Scale needs repeat of 15% or more on both cohorts
- And contribution of THB 3.51 or more per cube
- National listing is read at about M13

Desk assumptions. Cost per click THB 11.5 (proxy, one agency). Cost per thousand impressions THB 120 is an assumption with no saved source.

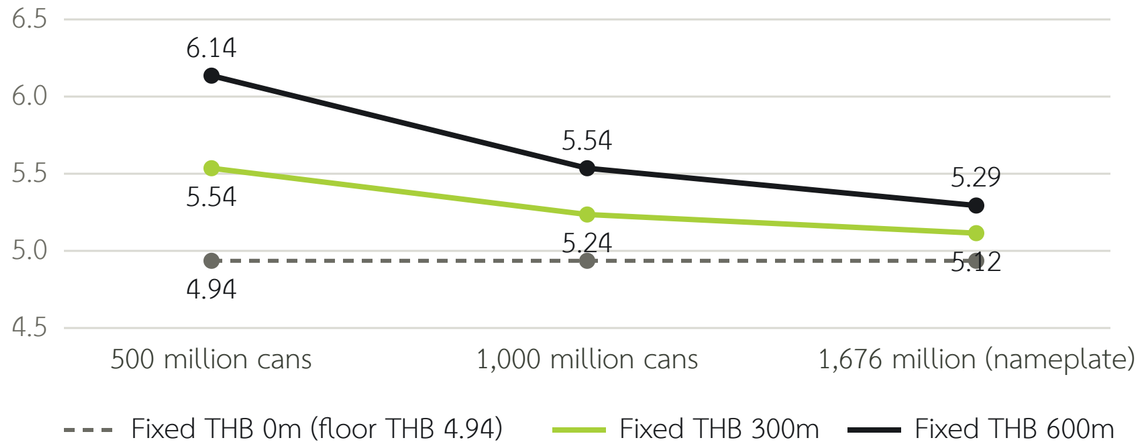
UNIT ECONOMICS

Each cube costs THB 4.09 and contributes THB 4.91 at a THB 9.00 planning price

THE CUBE: THB PER CUBE, SIX-CUBE PACK

Reactants	0.12
Pouch and print	2.80
Processing	0.50
Carton share	0.67
Cost per cube	**4.09**
Planning price	**9.00**
Contribution	**4.91**

THE CAN: THB PER CAN BY ANNUAL VOLUME AND FIXED COST



Desk estimate; no Singha or Boon Rawd can cost was found. Excludes excise, freight, distribution, marketing, overheads and capex. Scrap credit not netted, so cost is overstated. Inputs: LME aluminium USD 3,013 to 3,014 a tonne; THB 33.35 to 33.75 per USD (DERIVED).

Binder, filler and lemon flavour are not yet priced and are excluded.

Pouch THB 1.30, print THB 1.50 and processing THB 0.50 are assumptions. No Thai quote yet.

Planning price is 60% of a THB 15 shelf price (assumption). Contribution is 54.6% of the THB 9.00 planning price.

Desk model on assumed inputs. Reactants use import unit values and a sucralose price that is not yet verified.

DESK VALUATION

At 200,000 cubes a year and THB 500,000 fixed cost, the desk NPV is THB 0.88 million

THB 0.88m
NPV at 10%, base case

THB 0.68m
Maximum annual fixed cost for NPV = 0

146,230
cubes a year for NPV = 0 at THB 500,000 fixed cost

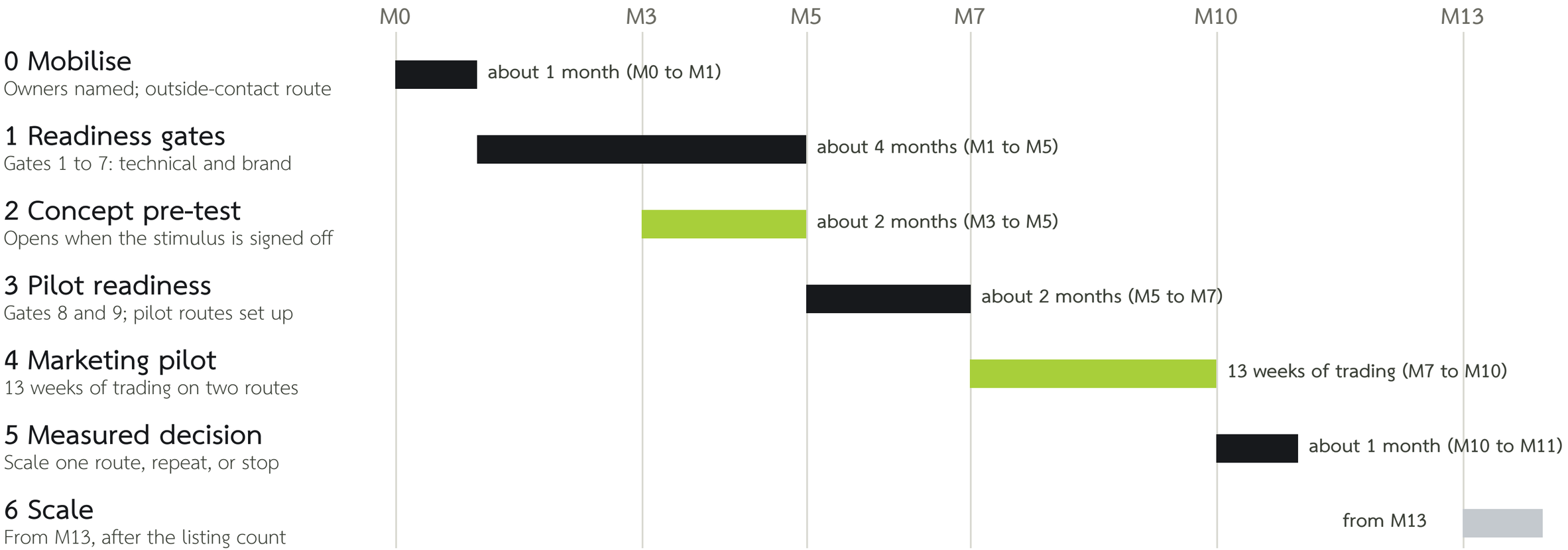
- At THB 250,000 fixed cost, NPV is THB 2.08 million.
- At THB 1,000,000 fixed cost, NPV is minus THB 1.51 million.
- At 50,000 cubes a year, NPV is negative at every fixed cost in the grid.
- Payback at THB 500,000 fixed cost is about 2.1 years.

Volumes and fixed costs are assumptions, not a forecast. Full grid in appendix B. The model excludes binder, flavour, excise (not yet quantified), VAT, tax, and marketing and pilot costs.

Desk scenario: five years, 10% discount rate, no terminal value, pre-tax. Year 0 carries one year of fixed cost as a set-up outlay, and year 1 sells half the base volume (both ASSUMPTION). Base volume 200,000 cubes a year is an assumption.

IMPLEMENTATION

The plan starts at Singha's go decision and runs in phases; nothing is dated until Singha sets M0



Gate 0 names owners within four weeks of M0. Sale waits for Gates 1, 2, 7 and 9.

Durations are assumptions for Singha to set. The storage test (Gate 5) starts by M2; its duration is an assumption.

GATES

Gates 1 to 9 stand between the desk model and a pilot batch; Gate 0 names the owners

Gate	Must show	Pass test	Owner (role)
0 Go and owners	Go decision; sponsor; one owner per gate	Named within four weeks of M0	Executive sponsor
1 Class and sweetener	Thai regulatory opinion; condition 127; annex edition check	Class settled in writing	Regulatory affairs
2 Potassium	Permission for potassium bicarbonate; medical view on 317 mg, covering people on potassium-retaining medicines or with kidney disease	Permission confirmed; adviser does not reject the serve	Regulatory affairs, medical adviser
3 Dissolved CO_2 (M2 to M5)	Bench cubes at 0.55 volume by M2; restated pass rule	Read against the rule R&D fixes before the run	R&D lead
4 Taste	Blind test at pH 3.5 and either side; lemon soda as reference	At least 55% blind preference (assumption)	R&D, consumer insight
5 Pack, storage, quotes	Film and seal spec; storage test from M2; quotes after approval	Reaction at most 0.5% of bicarbonate at 35 C; contribution at least THB 3.51 (ASSUMPTION)	Packaging R&D, procurement
6 Route and food safety	Named plant or co-packer; certificate verified for that site	One route with a verified certificate	Operations director
7 Brand, name, copy	Lion and SINGHA sign-off; alcohol-cue rulings; stimulus by M3	Written sign-off before any artwork or media	Singha brand and legal
8 Pilot design, budget	Routes, stores, duration; one budget; thresholds	Each parameter approved by Singha	Sales and channel, finance
9 Pack release	Weighed net weight; shelf life from the storage result	Printed only from weighed, tested samples	Packaging R&D, regulatory affairs

Sale is blocked until Gates 1, 2, 7 and 9 pass. A pilot batch also needs Gates 3 to 6 and 8. Thresholds marked assumption are for Singha to accept or change.

STOP AND SCALE

Stop rules are set in advance; a dose or acid failure gets one rework, then stops

PASS

Move to the next gate

FAIL ONCE

Dose or acid reworked once inside the window, and written down

FAIL TWICE

Stop that format

Trigger	Threshold	Action
Sweetener	No permitted sweetener for the class	No sale
Taste	Blind preference below 55% at pH 3.5	One acid reset, then stop
Pack	In-pack reaction above 0.5% of bicarbonate at 35 C	No launch until the barrier changes
Economics	Contribution below THB 3.51 per cube, or repeat below 15% on two cohorts	Stop scale-up
Food safety †	No verified certificate for the named site	No pilot batch
Velocity †	Below 3 units per store per week on two reads	Stop expanding that route
Paid social †	No cost-per-first-purchase threshold set	Route B does not start

† Proposed for Singha's written sign-off. Sell-through below 60% in 90 days is a review, not a stop.

DECISIONS FOR SINGHA

Twelve decisions from Singha start the work; Decisions 1 and 3 open the first phases

M0

1 Go decision and owners

Approve the plan as the basis for a go; name the sponsor and one owner per gate.

M1

2 Product class

Choose beverage, edible ice or electrolyte drink; commission the Thai opinion.

M1

3 Outside contact

Approve each contact separately, with a named provider and budget.

M1

4 Bench cubes, CO_2 method

Where the bench cubes are made; the pass rule; the 0.55 dose after one rework.

M2

5 Dry-solids route

Own line, co-packer or pilot run; the filler and the legal entity.

M2

6 Sold unit and carton

Confirm one cube per pouch; six, twelve or twenty-four cubes per carton.

M2

7 Brand and legal rulings

The lion and SINGHA beyond this proposal; alcohol cues; the name.

M4

8 Pilot routes

Singha chooses the pilot routes, the store count and location, the duration and the volume.

M4

9 One budget

One campaign and pilot budget, in one currency unit.

M4

10 Paid-social threshold

Cost per first purchase, cohort windows and the cannibalisation response.

M4

11 Traffic for the read

Existing channels, in-store or a listing, for the click-to-purchase read.

M3

12 Segment reading

Which direction of result keeps men aged 25 to 34 as the lead.

Full decision list, owners and the units each one unblocks: section 10 of the written proposal.

Thank you.

We are ready for questions.



Lemon Soda · Proposal to the SINGHA Ultimate Challenge 2026

APPENDIX A

What the label may say, and what it may not

May say

- Composition as fact: potassium bicarbonate, citric acid, sucralose and lemon flavour; no sodium carbonate. The binder and filler are not yet named or checked for sodium.
- "Drop one cube into 330 ml of water and stir"
- "Single-serve pouch"
- "Contains an intensive sweetener: sucralose"
- Net weight only after weighing

Barred, with no clearance route

- Sugar-free, zero-sugar and no-added-sugar wording
- Sodium-free and low-sodium wording
- Health, function, electrolyte and hydration wording
- Vitamin, potassium-benefit and calorie wording
- Certification and award wording; none is verified

Held until Gate 7 or a Thai legal review

- Novelty words
- Fizz, sparkling, bubbles and CO_2 release, until a measured test exists
- Lemon fruit imagery; glassware and bottle shape
- Alcohol cues
- The lion or SINGHA without written sign-off

Every consumer-facing word needs legal and brand approval before any stimulus, sampling or pilot print.

APPENDIX B

Valuation grid: NPV by volume and fixed cost (THB million)

Cubes a year	Fixed cost THB 250k	THB 500k	THB 1.0m	THB 2.0m	Max fixed cost, NPV = 0
50,000	(0.38)	(1.58)	(3.97)	(8.76)	0.17
200,000 (base)	2.08	0.88	(1.51)	(6.31)	0.68
1,000,000	15.18	13.99	11.59	6.80	3.42
Cubes a year for NPV = 0 (base case)	**73,115**	**146,230**	**292,461**	**584,922**	

Five-year horizon, 10% discount rate, no terminal value, pre-tax; year 0 carries one year of fixed cost, and year 1 sells half the base volume (both ASSUMPTION). Negative values are in brackets. Contribution is THB 4.91 per cube on the six-cube pack; volumes and fixed costs are assumptions.

Desk model, not a forecast. Single-factor sensitivities (shelf price, ex-factory share, pouch, print, carton) are in the written proposal, section 05.

APPENDIX C

Sources and evidence tiers

EVIDENCE TIERS

P	Regulator text read in full, company release, official statistics
S	Press, consultancy page, reproduced handbook, search snippet
UNVERIFIED	Not read, blocked, or a snippet only
PROXY	Stands in for a value that has not been measured
ASSUMPTION	Chosen for the model; not sourced
DERIVED	Our arithmetic, from inputs shown

PRINCIPAL SOURCES (ACCESSED OCTOBER 2026 UNLESS DATED)

- S · Singha Lemon Soda product page, extract (company says no sugar on three flavours; sucralose and acesulfame potassium): singhacorporation.com/singhalemonsoda/th/
- P · Thai food-additive annex, Notification 468, flavoured-drink rows (edition check open): food.fda.moph.go.th
- P · Royal Gazette, B.E. 2560, sugar-drink excise item 02.02: ratchakitcha.soc.go.th
- P · Bank of Thailand USD rate, 15 October 2026: bot.or.th/en/statistics/exchange-rate.html
- P · Ricola Drink Cubes release, 5 June 2025: ricola.com
- P, unverified page · NSO Social Indicators 2025, Table 4.6.2: nso.go.th
- S · CO_2 solubility, Lange attributed: bionumbers.hms.harvard.edu/bionumber.aspx?id=106208
- S · Thai potassium reference 3,500 mg (secondary copies; primary not opened)
- S · Container specifications, ONE and SeaCube: one-line.com; seacubecontainers.com

The full source list, with dates and the tier of each figure, is in section 10 of the written proposal.

UNVERIFIED figures (pouch listing, sucralose price, Thai margins, cost per impression) are desk-model inputs, not measured results; the unit economics on page 12 depend on them.